1. Definition

Data representation: The method of encoding information (numbers, text, images, sound) into a form that a computer can process, store, and manipulate—usually binary (0s and 1s).

2. Analogue vs Digital

(a)

- **Analogue data**: Continuous values (e.g., sound waves, temperature).
- **Digital data**: Discrete values represented in binary (0s and 1s).

(b) Sketches:

- *(i) Analogue signal*: Smooth, continuous sine-like wave.
- *(ii) Digital signal*: Square, step-like pulses (high/low).

3. Popularity of Binary

- Simple hardware implementation (two states: ON/OFF).
- High reliability against noise/errors.

4. Role of Modem

A **modem** converts digital signals from a computer into analogue signals for transmission over telephone lines, and vice versa, enabling communication.

5. Distinctions

(i)

- **Byte**: 8 bits.
- **Nibble**: 4 bits.

(ii)

- **Word**: Fixed-size unit of data handled by CPU (e.g., 16, 32, or 64 bits).
- **Word length**: Number of bits in a word.

6. Ascending Order

BIT → NIBBLE → BYTE → FILE

7. Number Systems

- **A**: Binary (0,1)
- **B**: Octal (0–7)
- **C**: Decimal (0–9)
- **D**: Hexadecimal (0–9, A–F)

8. Arithmetic Conversions

(a) Decimal → Binary

- $11 \rightarrow 1011$
- $1 \rightarrow 0001$
- $457 \rightarrow 111001001$

(b) Octal → Binary

- $77_8 \rightarrow 111111_2$
- $1_8 \rightarrow 000001_2$

(c) Binary Addition

- 4_{10} (100) + 3_{10} (011) = 111_2 (7_{10})
- 13_{10} (1101) + 2_{10} (0010) = 1111_2 (15_{10})

(d) Hex → Binary

- $C3_{16} \rightarrow 1100\ 0011_2$
- $13_{16} \rightarrow 0001\ 0011_2$

(e) Binary → Hex

- $110111.11_2 \rightarrow 37.\ C_{16}$
- $1.1110101_2 \rightarrow 1.\ E5_{16}$
- $1100001111111111112 \rightarrow C3FFF_{16}$

9. Hexadecimal

(a) Used for memory addresses, machine code, and color codes. **(b)** $767_8 \rightarrow 3FF_{16}$

10. One's Complement Subtraction

- $9 – 6 = 3$
- $17 – 15 = 2$
- $1110_2 – 1011_2 = 0011_2$ (3)
- $111010_2 – 110011_2 = 111_2$ (7)

11. Conversions

- $20.2_{16} \rightarrow 32.125_{10}$
- $11101_2 \rightarrow 29_{10}$

12. Binary Arithmetic

(a) 75 + 45 = 120 → 1111000_2 **(b)** Two's complement subtraction:

- 10111 – 10001 = 0110 (6)
- 11000 – 10011 = 0101 (5)

13. Two's Complement (Decimal)

- 23 – 20 = 3
- 17 – 14 = 3

14. Binary Arithmetic

- 11100111 + 00101110 = 100011101
- 1000 – 101 = 0011 (3)

15. Decimal Fraction

$4¾ = 4.75 \rightarrow 100.11_2$

16. BCD → Hex

10001001_2 (BCD) → 89_{16}

17. Two's Complement (8-bit)

-210 → 11111110_2

18. Hex → Binary

$FC1_{16} \rightarrow 1111\ 1100\ 0001_2$

19. Hex → Decimal

$7AE_{16} = 1966_{10}$

20. Methods of Binary Representation

- **Binary coded decimal (BCD)**
- **Gray code**
- **ASCII/Unicode (character codes)**

21. BCD

(a) Represents each decimal digit separately in 4-bit binary. **(b)** $451_{10} \rightarrow$ 0100 0101 0001 (BCD)

22. Mixed Conversions

(a) $1001_2 - 0111_2 = 0010_2$ (2) **(b)** 4 – 7 (two's complement) = -3 **(c)**

- $91B_{16} \rightarrow 4433_8$
- $376_8 \rightarrow FE_{16}$
- $9.625_{10} \rightarrow 1001.101_2$